

Observations on phytoplankton bloom dynamics during winter monsoon in Arabian Sea: a comparative analysis using Oceansat-3 OCM and MODIS Aqua Satellite data

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The Arabian Sea is among the most productive basins in the global ocean, and it is important in the context of reversal of monsoon, eddies, convection processes, winter cooling and significant upwelling phenomena. These processes influence the biogeochemical properties of the region. The formation of a phytoplankton bloom in the Arabian Sea during the winter monsoon is an example of biophysical coupling. Low sea surface temperature (SST) and high nitrate concentration result in enhanced concentration of chlorophyll-*a* (Chl-*a*) which is considered as the signal of phytoplankton bloom. The present study is carried out using Oceansat-3 Ocean Colour Monitor (OCM) and Moderate Resolution Imaging Spectroradiometer (MODIS) Aqua derived Chl-*a*, SST, photosynthetically active radiation (PAR), and modelled nitrate nutrient image analysis to study details of phytoplankton bloom dynamics on a weekly and monthly scale. The study has been carried out using the satellite data during the winter monsoon (November 2023 to February 2024) using 8-day composite/weekly-scale data and respective monthly data. Chl-*a* concentration ranged between 0.3 to 1.1 mg m⁻³, SST ranged 26.98°C–29.38°C, PAR ranged 35.87–50.26 Einstein m⁻² day⁻¹, nitrate ranged 2.3–4.1 μmol/L during the winter monsoon. The peak bloom condition is observed in the month of February during the winter monsoon. But PAR is observed to be increased in the peak bloom stages. Nitrate is the major controlling factor for the formation of bloom as compared to other parameters. OCM-3 data is showing matching results with MODIS data with little overestimation. The present study proves the suitability of the Oceansat-3 OCM sensor data for bloom observation in the Arabian Sea during the winter monsoon from its initiation phase to its peak phase and for understanding the dynamics and correlation of Chl-*a*, SST, PAR, and modelled nitrate

with phytoplankton biomass variability in bloom and non-bloom sites.

Keywords: Arabian Sea, chlorophyll, MODIS-Aqua, Oceansat-3 OCM, phytoplankton bloom.

THE Arabian Sea (AS) is one among the highly productive regions in the tropical oceans, and its boundary and open ocean processes are impacted by summer upwelling and winter cooling, which introduces large amounts of nutrients into the upper ocean, improving primary production and eventually the fisheries¹. In recent years, the phytoplankton blooms are observed at regular intervals and are being reported in the Arabian Sea, mainly occurring during winter monsoons^{2,3}. Such blooms tend to increase in species cell density over the time period. Phytoplankton consists of small, single-celled organisms, which constitute the base of the oceanic food web. The abundance of phytoplankton in the oceans is controlled by biophysical processes and anthropogenic and climate-induced variables. Hence, the phytoplankton growth, which leads to a bloom, is triggered by spatiotemporal variation in water quality parameters^{4,5}. The spatio-temporal observations of phytoplankton blooms on a synoptic scale have been possible through space technology via satellites with larger area coverage⁶. The *in situ* based measurements are not cost-effective and have limitations to cover a broad area in a short time span. Remote sensing observations have been vital to estimate and monitor important ocean parameters such as chlorophyll-*a* (Chl-*a*) concentration, phytoplankton absorption, sea surface temperature (SST) and photosynthetically available radiation (PAR) over large areas. Satellites are capable of estimating various parameters in surface waters on a synoptic scale by measurement of spectral changes of the upwelling light. Satellite sensors observe and quantify the colour of the ocean surface, providing valuable information about various parameters and processes. This approach, known as ocean colour remote sensing, measured by satellites, unveils chlorophyll content, a key proxy for phytoplankton biomass. Ocean colour satellite sensors like the Sea-viewing Wide Field-of-View Sensor (SeaWiFS), Moderate Resolution Imaging Spectroradiometer (MODIS)-Aqua, Ocean Colour Monitor (OCM), Visible Infrared Imaging Radiometer Suite (VIIRS), etc. have successfully mapped and visualised the phytoplankton biomass in terms of Chl-*a* concentration on a regional and global scale⁷. Oceansat-3 satellite configured and launched into orbit to measure physical, biological and atmospheric parameters. Its sensor, OCM-3, is designed for providing quantitative information of ocean colour variables, e.g., Chl-*a* concentration, total suspended matter, etc. It is operating in two modes: (i) local area coverage at 366 m resolution and (ii)

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global area coverage mode at 1.1 km resolution, providing observations in 13 narrow (10–20 nm) spectral channels in visible, near-infrared and short-wave infrared ranges (400–1030 nm). A significant improvement in bandwidth (10–20 nm) from its predecessor, OCM-2, having 8 wider bands, 20–40 nm bandwidths⁸. OCM-3 being a global sensor provides high-resolution chlorophyll products for global ocean colour applications.

The MODIS-Aqua sensor views the entire Earth's surface every 2 days, acquiring data in 36 spectral bands covering the range from 400 nm to 14.4 μ m with 4 km resolution, and its bands of primary interest to ocean colour applications are the 9 channels covering the spectral range from 400 nm to 900 nm (ref. 9). The present study provides an insight into the recent phytoplankton bloom phases in the northern Arabian Sea and their dynamics using two different sensors in space orbit for the retrieved data and their linkage to relevant corollary surface ocean parameters.

Methodology

Study area

The study area selected for the present study is the Arabian Sea (55°–78° E and 6°–24° N). Data were acquired for the period of November 2023 to February 2024 (winter monsoon). A phytoplankton bloom was recorded in irregular patches based on chlorophyll concentrations in the subregions of the northern Indian Ocean (21°–23° N and 63°–65° E) and observations were made in non-bloom subregions (16°–19° N and 68°–70° E) for the comparative observations on prevailing environmental conditions (Figure 1).

Results and discussion

Satellite data

For the estimation of phytoplankton pigment variability in the Arabian Sea, data were acquired from two satellites. The Indian remote sensing satellite Oceansat-3 OCM Chl-*a* concentration level-2 data with a spatial resolution of 1 km were downloaded from the Bhoonidhi Portal (nrsc.isro.gov.in) for the period November 2023 to February 2024. The dates of access of OCM-3 data from the National Remote Sensing Centre-Indian Space Research Organization (NRSC-ISRO) Bhoonidhi Portal were 10 March 2024, 20 March 2024 and 12 April 2024. The OCM-3 on-board Oceansat-3 has 13 narrow (10–20 nm) spectral channels with 2 days' temporal resolution.

MODIS Aqua level-3 Chl-*a*, SST and PAR data during November 2023 to February 2024 were downloaded from ocean colour, National Aeronautics and Space Administration (NASA) (<http://modis.gsfc.nasa.gov>), with

a spatial resolution of 4 km. Data products such as Chl-*a* and SST were used as proxies for the estimation of sea surface nitrate.

Data processing

Processing and analysis of the Oceansat-3 OCM and MODIS-Aqua data have been carried out using SeaWiFS Data Analysis System (SeaDAS) version 9.0 software. The SeaDAS, a comprehensive image analysis software developed and maintained by NASA (<http://seadas.gsfc.nasa.gov>) was used for processing, displaying, analysing and quality controlling data related to ocean colour. Mosaicking and resampling of OCM-3 data had been done to get the distribution of Chl-*a* in the entire Arabian Sea to compare with MODIS data.

Results and discussion

Phytoplankton pigment variability

To substantiate the trend of variation in Chl-*a* during the winter monsoon, data were acquired from MODIS during four seasonal years (2020–2024). The bloom indicating high Chl-*a* was recorded during the month of February (Figure 2).

Figures 3 and 4 have shown the variation of Chl-*a* (8-day composite) from OCM-3 and MODIS, respectively, in the Arabian Sea during the winter monsoon. Chl-*a* concentration observed during the winter monsoon ranged between 0.1 to 2.0 mg m⁻³ in the Arabian Sea. OCM-3 data in Figure 5 shows matching results with MODIS data, even if it is overestimated. This indicates the suitability of the Oceansat-3 OCM sensor data for bloom observation in the Arabian Sea during the winter monsoon from its initiation phase to peak phase. The winter cooling phenomenon in the Arabian Sea is caused by the northeast monsoon winds impact on dry, cool continental air, which increases evaporation and causes the water's surface to cool^{10,11}. The solar insolation gets reduced during January–February in the Arabian Sea, which induces the increase in mixed layer depth. Concurrently, the densification and cooling observed in the northern Arabian Sea are a winter effect; this process leads to sinking and convective mixing. Subsequently, the water column nutrients are injected into surface layers from thermocline¹². As understood, the detrainment also played a key role in thinning the mixed layer and enabling more light availability for phytoplankton growth, favouring blooms¹³. However, these systems are typically short-lived, as detrainment fails to replenish nutrients in the mixed layer, limiting their persistence to the duration of the initial nutrient availability. Detrainment blooms also die due to nutrient constraint, although the first peak

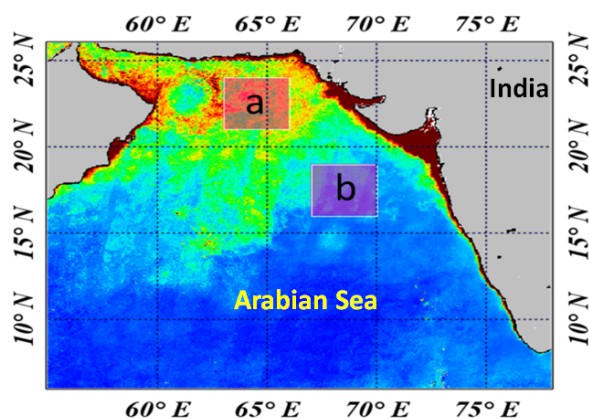


Figure 1. Map showing the study area. Sub regions (a) Showing the bloom area, and (b) Showing the non-bloom area.

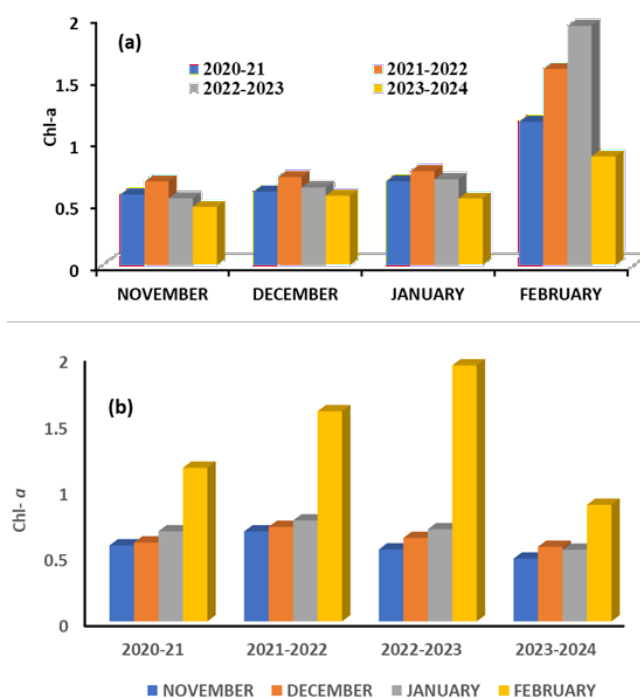


Figure 2. (a) and (b) Bar diagram showing variation of chlorophyll-a (Chl-a) in the Arabian Sea during the winter monsoon (2020–2024).

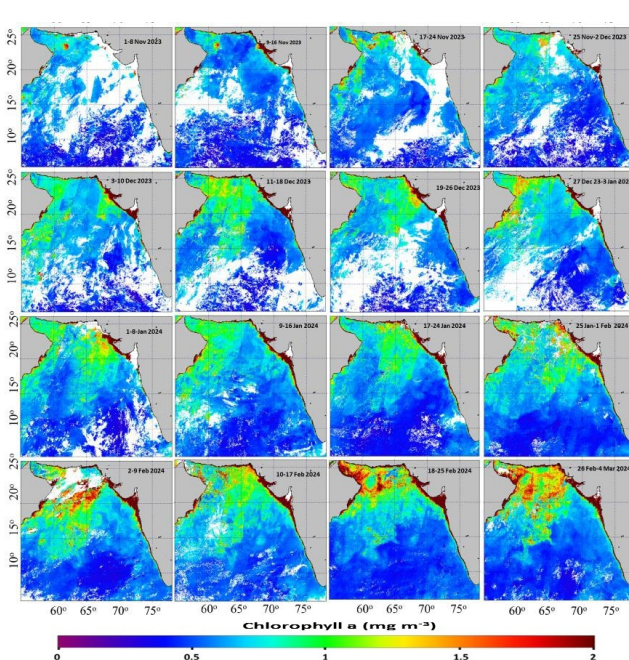


Figure 3. Variation of Chl-a in the Arabian Sea during the winter monsoon (Oceansat-3, 8-day composites).

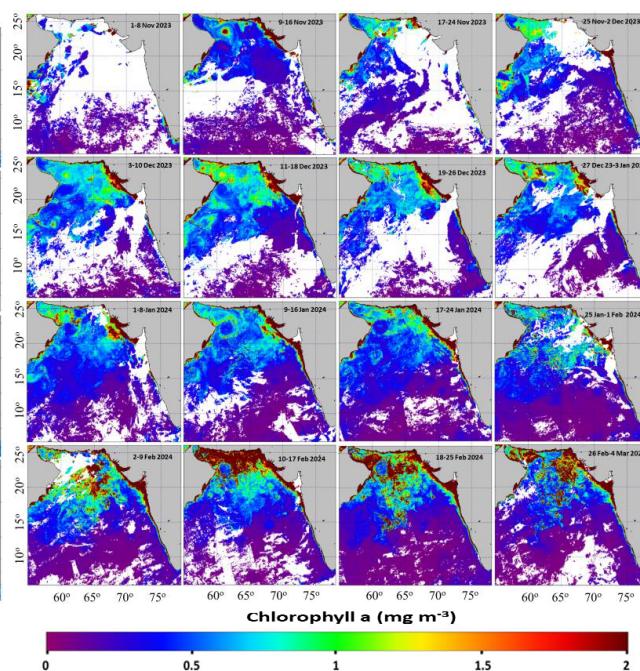


Figure 4. Variation of Chl-a in the Arabian Sea during the winter monsoon (Moderate Resolution Imaging Spectroradiometer (MODIS) 8-day composites).

decays quickly due to self-shading and grazing. Grazing has been an important process restricting the magnitude

of phytoplankton biomass and concentration in different bloom stages, from peak to lean phases.

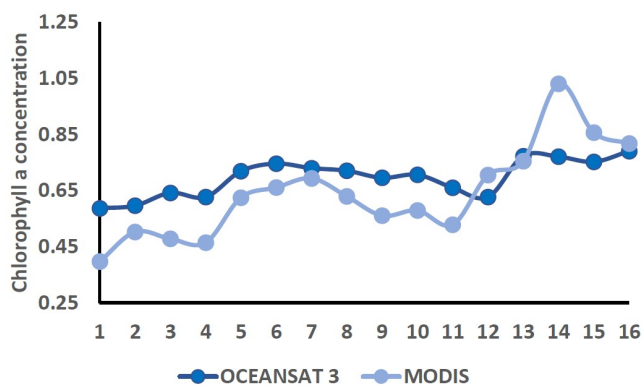


Figure 5. Graph showing variation of Chl-*a* in Arabian Sea using Oceansat-3 and MODIS 8-day data during November 2023–February 2024.

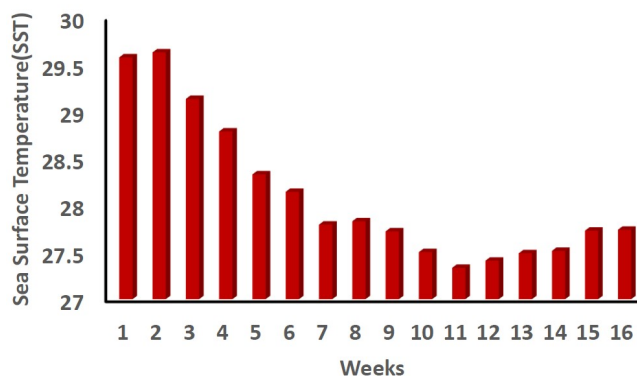


Figure 7. Bar diagram showing variation of SST in the Arabian Sea during winter monsoon (8-day composite).

Impact of SST on phytoplankton bloom

To analyse the impact of SST on variation in chlorophyll concentrations in the Arabian Sea, MODIS 8-day and monthly images are processed (Figure 6). From November 2023 to February 2024, SST decreased from 29.38°C to 28°C (Figure 7).

Interestingly, a significant negative association was observed between SST and chlorophyll *a* concentration. In the present study, Oceansat-3 and MODIS showed an inverse relationship between chlorophyll and SST with R^2 values of 0.457 for OCM-3 and 0.3975 for MODIS. The distribution of Chl-*a* and SST are inversely correlated (Figure 8). The relationship between SST, chlorophyll concentration, and phytoplankton blooms in the Arabian Sea is complex and interconnected. The increased concentration of chlorophyll during the winter months is caused by mesoscale eddies and winter convective mixing that occurred due to wind stirring, which raise the mixed layer of rich nutrients to the surface^{14,15}. The enrichment of

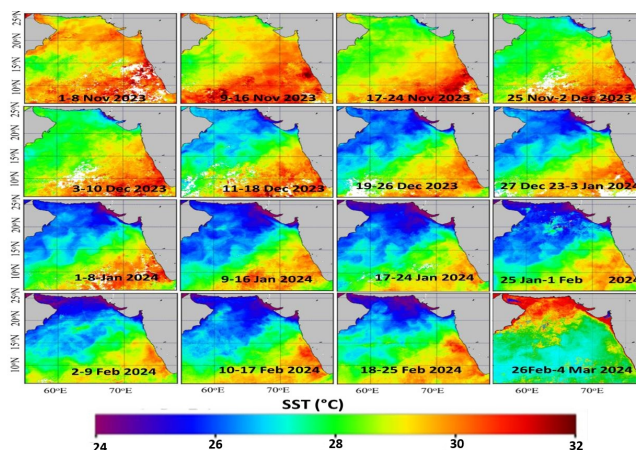


Figure 6. Weekly scale (8-days composite interval) variability of sea surface temperature (SST) in the Arabian Sea during winter monsoon.

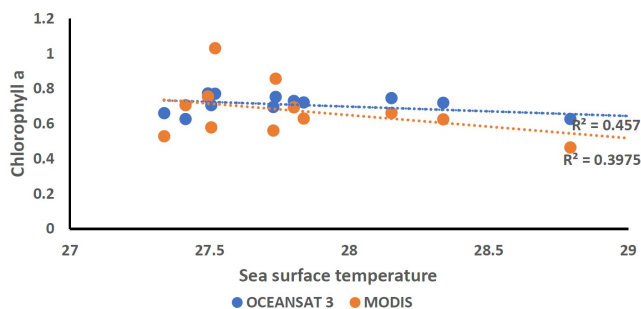


Figure 8. Relation between Chl-*a* and SST.

the mixed layer through cool thermocline water and this cooling event is responsible for the increase in surface chlorophyll concentrations¹⁶. The phytoplankton in denser concentrations have been able to trap shortwave solar radiation and its absorption, which increased the SST¹⁷.

Impact of PAR on phytoplankton bloom

Figure 9 shows variation of PAR in the Arabian Sea during the winter monsoon, 16 weeks during November 2023–February 2024. PAR is observed to be increased in the peak bloom stages (Figures 10 and 11). PAR ranged from 35.877 to 50.265 Einstein $\text{m}^{-2} \text{day}^{-1}$ during winter monsoon. PAR is enhancing phytoplankton bloom up to a limit. Phytoplankton blooms in the Arabian Sea are strongly influenced by the amount of PAR reaching the surface waters. PAR is the specific range of wavelengths of sunlight that phytoplankton can use for photosynthesis, the process by which they capture energy to grow and reproduce. PAR is related to phytoplankton

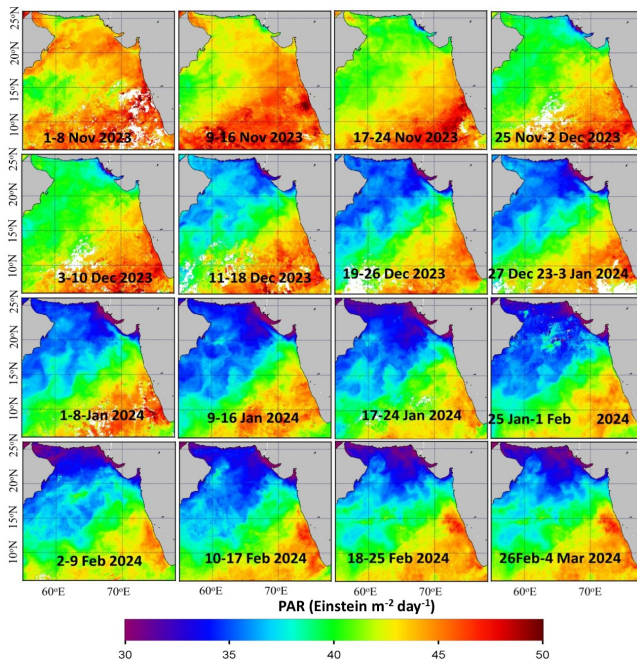


Figure 9. Variation of photosynthetically available radiation during winter monsoon (16 weeks during November 2023–February 2024).

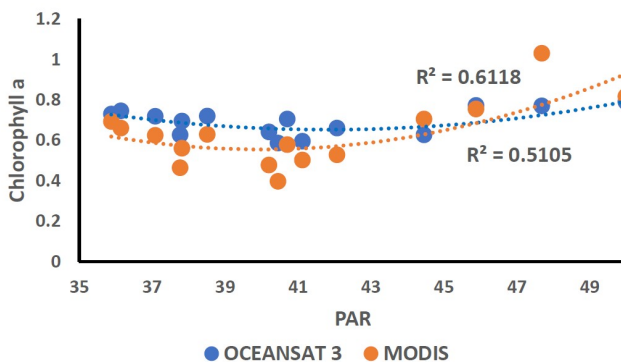


Figure 11. Relation between Chl-*a* and PAR (Oceansat-3 and MODIS).

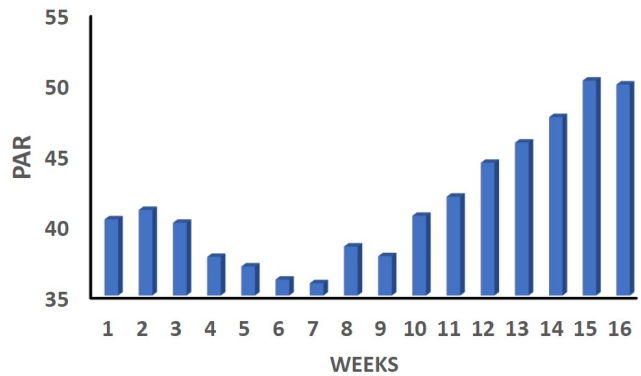


Figure 10. Graph showing variation of photosynthetically active radiation (PAR) in Arabian Sea during winter monsoon (16 weeks during November 2023–February 2024).

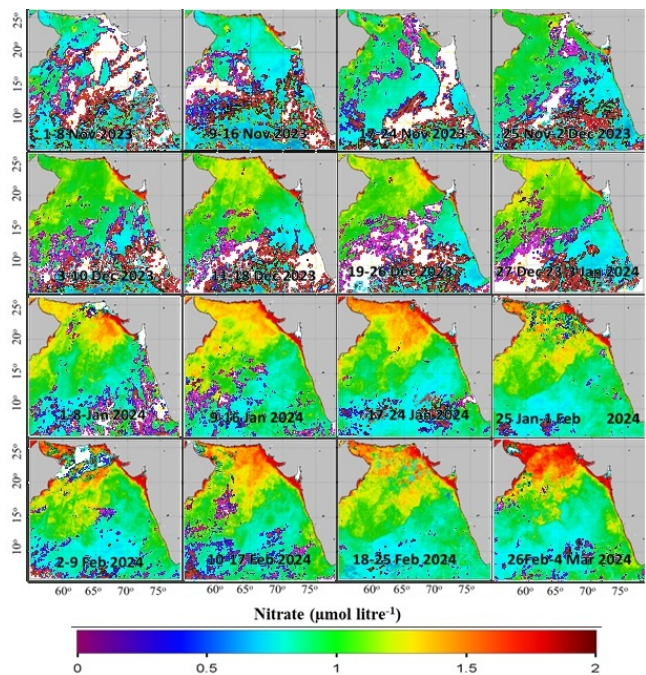


Figure 12. Variation of nitrate in Arabian Sea during the winter monsoon (with input of Oceansat-3 and MODIS Chl-*a* data).

blooms in the Arabian Sea. Phytoplankton need sufficient PAR for photosynthesis. During the monsoons (both summer and winter), wind patterns cause mixing of the

water column, bringing nutrients from deeper layers to the surface. This upwelling creates ideal conditions for phytoplankton growth, as long as there's also sufficient

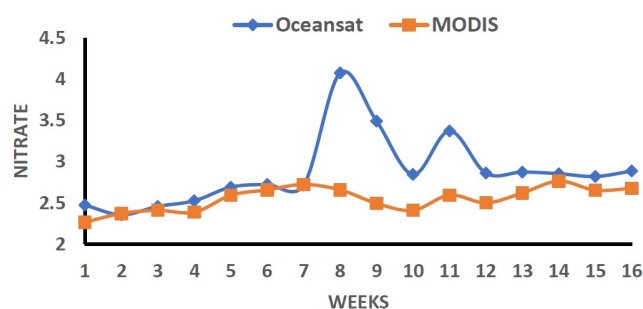


Figure 13. Graph showing variation of nitrate in Arabian Sea during winter monsoon.

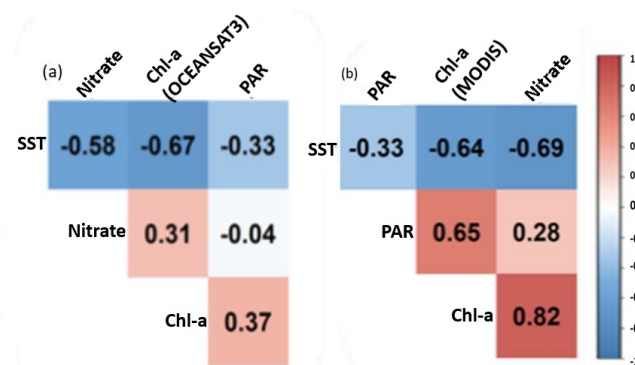


Figure 14. Correlation analysis of different parameters which influencing phytoplankton bloom in entire Arabian Sea with (a) Oceansat-3 (b) MODIS-Aqua.

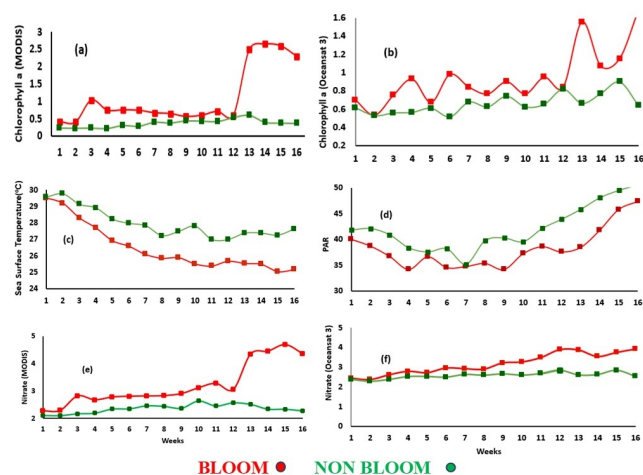


Figure 15. (a) Variation of Chl-a from MODIS, (b) Chl-a from Oceansat-3, (c) MODIS SST, (d) MODIS PAR, (e) Nitrate (MODIS), and (f) Nitrate (Oceansat-3) in bloom and non-bloom regions.

sunlight penetration. The area of the ocean where enough sunlight reaches to support photosynthesis is known as the euphotic zone.

The euphotic depth of the Arabian Sea is dependent on water turbidity and phytoplankton density, and these are connected to water clarity coincident with light availability. Higher PAR allows for a deeper euphotic zone, providing a larger volume for phytoplankton to thrive. Even with sufficient PAR, other factors like nutrient availability can limit the extent of a bloom. Certain nutrients, like nitrate and silicate, are crucial for phytoplankton growth. Different phytoplankton species have varying light requirements. Some may be more efficient at utilising lower levels of PAR compared to others. Overall, PAR acts as a key regulator for phytoplankton growth in the Arabian Sea, influencing the timing, intensity, and overall succession of blooms.

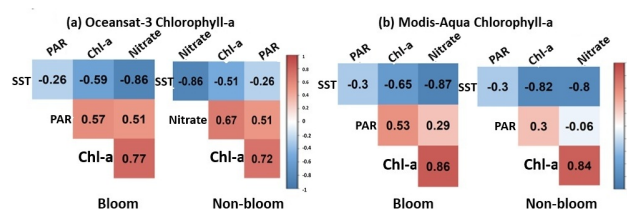


Figure 16. Correlation analysis of different parameters in bloom and non-bloom regions using (a) Oceansat-3 and (b) MODIS-Aqua data.

Variation of nutrients

Estimation of sea surface nitrate concentration from remotely sensed SST from MODIS Aqua and Chl-a, from both Oceansat-3 and MODIS data, during November 2023 to February 2024 has been studied. By using equation¹⁸ modelled nitrate image obtained (Figure 12). From Figure 13, Oceansat-3 modelled nitrate is indicating the productive zones variation in the Arabian Sea during the winter monsoon. The value ranges between 2.2 to 4.2 $\mu\text{mol/L}$ in the Arabian Sea during the winter monsoon. Nutrients are generally thought to be supplied to surface waters from the nutrient-rich deeper water by vertical diffusion or advection. Nitrate plays a critical role in fuelling phytoplankton blooms in the Arabian Sea, acting as a primary nutrient source (Figure 13). The winter cooling

and vertical mixing in the water column reduce the SST and increase the nutrient concentration in the late phase of the monsoon, exhibited by low light condition¹⁹. The nutrients (NO₃) and Chl-*a* have shown a significant positive correlation during the bloom incidence in the Arabian Sea (Figure 14).

Comparison between bloom and non-bloom region in Arabian Sea

Chl-*a* observed higher in the northern Arabian Sea compared to the southern Arabian Sea. The dynamics of physicochemical entities, nutrients and Chl-*a* variability were studied in the bloom and non-bloom locations along the Arabian Sea. Parameters enhancing phytoplankton bloom formation estimated in bloom and non-bloom areas during winter monsoon has been correlated. The chlorophyll, SST, PAR, and nitrate 8-day composites generated from the MODIS-Aqua and Oceansat-3 (EOS-06) OCM data acquired for the period November 2023 to February 2024. Correlation analysis of physicochemical, nutrient, and Chl-*a* variables was carried out for the bloom and non-bloom sites. For bloom and non-bloom regions in both satellites (MODIS and Oceansat-3) showing similar variation of Chl-*a* in the winter monsoon (Figure 15). The highest seasonal means and inter-seasonal variability of Chl-*a* observed in the western and northwestern Arabian Sea²⁰. Chl-*a* observed higher in the northern Arabian Sea compared to the southern Arabian Sea. The Arabian Sea's bloom and non-bloom sites were studied for the dynamics of physicochemical, nutritional, and Chl-*a* variables. Parameters enhancing phytoplankton bloom formation, estimated in bloom and non-bloom area during winter monsoon, correlated. Nitrate is the major controlling factor for the bloom occurrence (Figure 16) as compared to other parameters.

Conclusion

Phytoplankton blooms are frequent in the Arabian Sea. There are different parameters that impact the phytoplankton bloom. However, its impact and influence are complex and interrelated with various environmental factors like light availability, nutrients, and water column dynamics. Strong PAR coinciding with upwelling events during the monsoons is a key factor for triggering massive phytoplankton blooms in the Arabian Sea. The phytoplankton biomass in terms of chlorophyll variability has been a measure of SST and the thermal structure of the water column. Blooms can occur despite cooler SST compared to summer, as long as light availability is sufficient. Nitrate plays a critical role in fuelling phytoplankton blooms in the Arabian Sea, acting as a primary nutrient source. Nitrate is a vital

nutrient for phytoplankton blooms in the Arabian Sea. The Arabian Sea experiences blooms during winter monsoons due to upwelling events that bring deep, nutrient-rich waters to the surface. Oceansat-3 OCM data analysis inferred its suitability to observe phytoplankton blooms from initiation to peak phase as a proof-of-concept satellite launched by India with its indigenous development, having an excellent high resolution and advanced ocean colour sensor, OCM-3. Observations inferred that February is more productive during winter monsoon. More detailed studies, such as validation, nutrient ratio, and stoichiometry related to phytoplankton types and taxonomy, are required over the study area for further analysing phytoplankton bloom.

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